

Multi-Output Linear Regression: Linear $\rightarrow$ Multivariate Gaussian

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Abstract

The regression counterpart of softmax’s Bernoulli $\rightarrow$ multinoulli step: a vector response $y \in \mathbb{R}^K$ that is conditionally multivariate normal around a linear map of x . Unlike softmax, the closed form survives — the Frobenius loss separates over outputs into K ordinary least-squares problems sharing one Gram factorization — and this note records why, when correlated noise still doesn’t change the answer, and what the one-hot “linearization trick” for classification gets right and wrong. Demonstrations in `multioutput.py`.

1 The model

$$y \mid x \sim \mathcal{N}(W^\top x, \sigma^2 I_K), \quad W \in \mathbb{R}^{d \times K}, \quad (1)$$

one weight column per output. The conditional NLL, up to constants, is the Frobenius least-squares loss

$$\mathcal{L}(W) = \frac{1}{2n} \|XW - Y\|_F^2 = \sum_{k=1}^K \frac{1}{2n} \|Xw_k - Y_{:,k}\|^2, \quad (2)$$

and the second equality is the whole story: *the loss separates over output columns*. Each w_k solves an ordinary least-squares problem against the same design.

Proposition 1. *If X has full column rank, the unique MLE is*

$$\boxed{\widehat{W} = (X^\top X)^{-1} X^\top Y}, \quad (3)$$

the normal equations of the linear-regression note with K right-hand sides — one Gram factorization, reused for every output.

Proof. $\nabla_W \mathcal{L} = \frac{1}{n} X^\top (XW - Y) = 0$ gives $X^\top X W = X^\top Y$; column k of this system is exactly the single-output normal equation for $(X, Y_{:,k})$, with the usual strict convexity giving uniqueness. $\square$

Per sample, the gradient is $x_i (x_i^\top W - y_i^\top)$ — residual (now a K -vector) times features, the same GLM shape as everywhere else in this collection; the script’s SGD and autograd routes verify it by trajectory identity.

Remark 1 (Correlated noise changes nothing — here). *Replace $\sigma^2 I_K$ in (1) by a full covariance Σ : the NLL becomes a Σ^{-1} -weighted quadratic, yet its minimizer is still (3). Reason: the stationarity condition reads $X^\top(XW - Y)\Sigma^{-1} = 0$, and Σ^{-1} is invertible, so it drops out. This is the seemingly-unrelated-regressions collapse: correlated outputs only matter for estimation once the outputs use different regressors; with a shared design, per-column OLS is already the MLE. (Estimating Σ itself: the $1/n$ residual covariance, as in the Gaussian note.)*

2 The one-hot linearization trick

Multi-output regression tempts a shortcut for K -class classification: encode labels as one-hot vectors $Y \in \{0, 1\}^{n \times K}$, solve (3), classify by $\arg \max_k$ of the fitted scores. What this gets *right*: linear score functions and often reasonable accuracy (the script measures it a few points below the softmax fit on the same data). What it gets *wrong*: the scores are not probabilities — the script’s histogram shows fitted values well outside $[0, 1]$ (they need not sum to 1 either), because a Gaussian likelihood on $\{0, 1\}$ targets is misspecified. Squared loss also penalizes *over*-confident correct scores, dragging boundaries toward outlying but well-classified points. The proper likelihood for a categorical response is the multinoulli — softmax regression — which is exactly the difference between modeling $\mathbb{E}[Y|x]$ with the wrong noise and modeling $\Pr(Y|x)$ itself.

3 Experiment

$n = 3000$, $d = 3$, $K = 2$, noise $\sigma = 0.8$ (seed 7): the closed form recovers W^* to sampling accuracy; per-column OLS reproduces it column by column (asserted); GD converges to it; hand and autograd SGD trajectories coincide to 10^{-10} . The one-hot trick on 3-class softmax data classifies within a few points of the Bayes rule while its “probabilities” span roughly $[-0.4, 1.3]$ — linear scores, broken semantics.